

E. Jeffrey Samuels (192524412)

SET 3 — Employee Performance Evaluation

According to your experiment sheet, Set 3 uses **Employee Performance** data and asks for a line chart, department bar chart, scatter plot, and Tableau dashboard.

Dataset

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

AIM

To analyze employee performance data using **R programming** and visualize employee performance trends, department distribution, and the relationship between years of service and performance scores.

ALGORITHM

1. Start RStudio.
2. Create the employee performance dataset.
3. Store Employee ID, Department, Years of Service, and Performance Score.
4. Create a **line chart** to visualize employee performance.
5. Create a **bar chart** to show employees across different departments.
6. Create a **scatter plot** between years of service and performance score.
7. Calculate the correlation between years of service and performance score.
8. Add suitable titles and axis labels.
9. Display the graphs and results.
10. Stop.

R CODE

```
employee <- data.frame(  
  ID=c(1,2,3,4,5),  
  Department=c("Sales","HR","Marketing","Sales","HR"),  
  Years=c(5,3,7,4,2),  
  Performance=c(85,92,78,90,76)  
)
```

1. Line Chart

```
plot(employee$Years,employee$Performance,  
      type="o",  
      main="Employee Performance Trend",  
      xlab="Years of Service",  
      ylab="Performance Score")
```

2. Bar Chart

```
dept <- table(employee$Department)
```

```
barplot(dept,  
        main="Employees by Department",  
        xlab="Department",  
        ylab="Number of Employees")
```

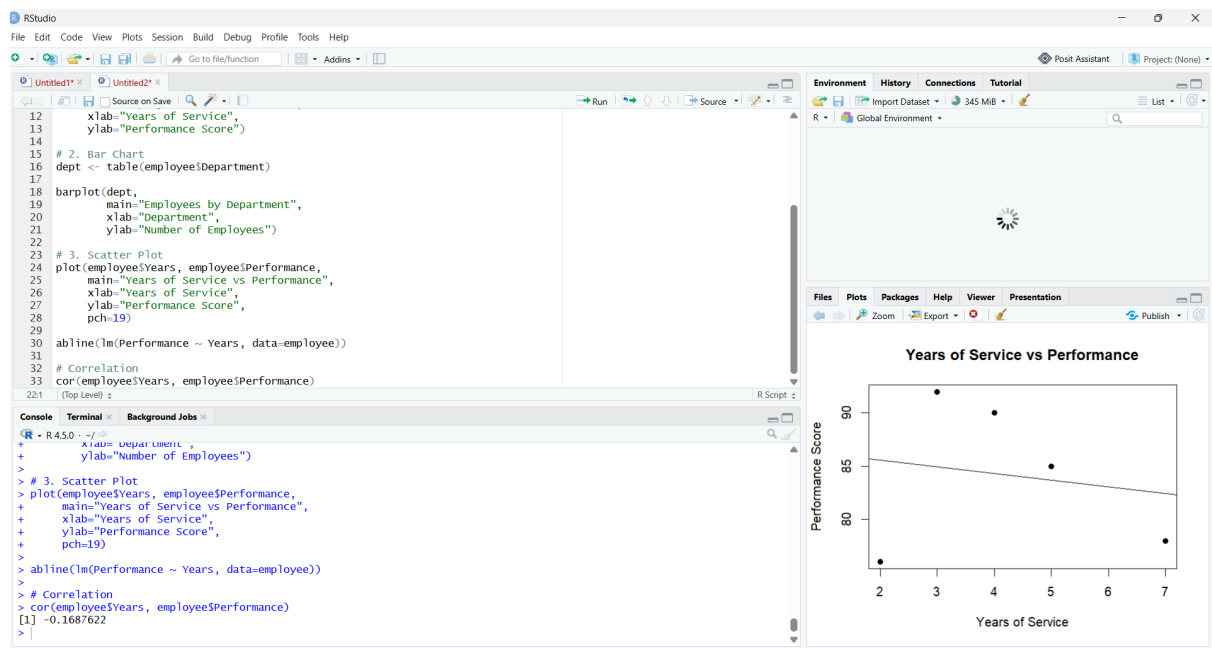
3. Scatter Plot

```
plot(employee$Years,employee$Performance,  
      main="Years of Service vs Performance",  
      xlab="Years of Service",  
      ylab="Performance Score",  
      pch=19)
```

```
abline(lm(Performance~Years,data=employee))
```

Correlation

```
cor(employee$Years,employee$Performance)
```



RESULT

Thus, the **Employee Performance** data was successfully analyzed using R programming. The **line chart, bar chart, and scatter plot** were successfully created to visualize employee performance, department distribution, and the relationship between years of service and performance score.